

# SkinSegAI: Technical Project Documentation

Few-Shot Medical Image Lesion Segmentation Using Transfer Learning with MobileNetV3-UNet

## 1. Executive Summary & Project Specifications

|                         |                                                                                                   |
|-------------------------|---------------------------------------------------------------------------------------------------|
| Project Title           | Few-Shot Medical Image Lesion Segmentation with Transfer Learning                                 |
| System Identifier       | SkinSegAI (Interactive Clinical Research Dashboard)                                               |
| Model Architecture      | MobileNetV3-Small (Encoder) + 4-Stage U-Net (Decoder with Skip Connections)                       |
| Core Technologies       | PyTorch 2.11, Torchvision 0.26, Streamlit 1.55, Plotly, NumPy, Pillow                             |
| Dataset Benchmark       | ISIC 2018 Task 1: Lesion Boundary Segmentation (Dermatoscopic RGB + Binary Masks)                 |
| Evaluated Splits        | 50, 100, 250, and 500 images (Low-Data / Few-Shot Regimes)                                        |
| Best Validation Metrics | Dice: 87.07%   IoU: 77.52% (at 500 images, Epoch 9)                                               |
| Remote Repository       | <a href="https://github.com/Gowthami1214/SkinSegAI">https://github.com/Gowthami1214/SkinSegAI</a> |

## 2. Clinical Background & Research Motivation

Malignant melanoma represents the most fatal form of skin cancer, yet early surgical excision yields a 5-year survival rate exceeding 99%. Clinical assessment relies on dermoscopy and the **ABCD criteria** (Asymmetry, Border irregularity, Color variegation, Diameter). Automated lesion boundary segmentation is the foundational first step: without an exact border, computerized systems cannot compute asymmetry, border irregularity, or pigment network distribution.

**The Data Scarcity Dilemma:** Standard deep learning segmentation models require thousands of pixel-level annotations. Manual pixel contouring by certified dermatologists takes 3–8 minutes per image, creating an insurmountable bottleneck for rare conditions or resource-limited health systems. SkinGenAI investigates: *What is the minimum labeled training set needed to produce clinically viable segmentation masks using transfer learning?*

## 3. Deep Learning Architecture: MobileNetV3-UNet

The model integrates a lightweight **MobileNetV3-Small** backbone (ImageNet-pretrained) with a symmetrical **U-Net decoder** via multi-scale skip connections. This design pairs high parameter efficiency (~1.5M parameters) with spatial boundary reconstruction.

| Pipeline Stage           | Feature Dim    | Channels       | Architectural Role                             |
|--------------------------|----------------|----------------|------------------------------------------------|
| Input Dermoscopy         | 224 × 224      | 3 (RGB)        | Standardized normalized dermoscopic photo      |
| Encoder Stage 0 (Skip 4) | 112 × 112      | 16             | Fine hair, edge, and contrast transitions      |
| Encoder Stage 1 (Skip 3) | 56 × 56        | 16             | Local pigment networks and texture maps        |
| Encoder Stage 2 (Skip 2) | 28 × 28        | 24             | Regional lesion symmetry features              |
| Encoder Stage 8 (Skip 1) | 14 × 14        | 48             | Lesion global shape context                    |
| Bottleneck Bridge        | 7 × 7          | 576            | Highest semantic abstraction                   |
| Decoder UpConv 1–4       | 7→14→28→56→112 | 48, 24, 16, 16 | ConvTranspose2d upsampling + Skip fusion       |
| Final Projection         | 224 × 224      | 1 logit        | Sigmoid → Binary thresholding at $\theta=0.50$ |

## 4. Optimization & Validation Metrics

**Hybrid Loss:** Dermoscopy images exhibit severe foreground/background class imbalance (lesions cover only 15–30% of canvas area). SkinGenAI trains with  $L_{total} = L_{BCE} + L_{Dice}$ , where BCE stabilizes per-pixel probability convergence and Dice Loss directly penalizes missed lesion areas.

| Metric | Formula | Clinical Interpretation |
|--------|---------|-------------------------|
|--------|---------|-------------------------|

|                       |                                       |                                                                                             |
|-----------------------|---------------------------------------|---------------------------------------------------------------------------------------------|
| Dice Similarity (DSC) | $2 \cdot TP / (2 \cdot TP + FP + FN)$ | Gold standard ISIC metric. Measures contour completeness and overlap harmonic mean.         |
| IoU (Jaccard Index)   | $TP / (TP + FP + FN)$                 | Strict overlap area divided by total union area. Mathematically equal to Dice / (2 - Dice). |
| Precision             | $TP / (TP + FP)$                      | Measures over-segmentation. High precision ensures minimal healthy skin is excised.         |
| Recall (Sensitivity)  | $TP / (TP + FN)$                      | Critical oncology metric. High recall guarantees malignant tumor cells are not missed.      |

5. Empirical Benchmark Results across Few-Shot Configurations

| Configuration n | Training Samples | Best Epoch | Validation Dice | Validation IoU | Empirical Progression                                     |
|-----------------|------------------|------------|-----------------|----------------|-----------------------------------------------------------|
| Config 1        | 50 images        | 15         | 0.5104 (51.0%)  | 0.3621 (36.2%) | Baseline: Identifies dark centroids; rough borders        |
| Config 2        | 100 images       | 15         | 0.5765 (57.6%)  | 0.4106 (41.1%) | +6.6% Gain: Learns radial symmetry & artifact suppression |
| Config 3        | 250 images       | 15         | 0.8246 (82.5%)  | 0.7081 (70.8%) | Critical Inflection Point (+24.8% jump)                   |
| Config 4        | 500 images       | 9          | 0.8707 (87.1%)  | 0.7752 (77.5%) | Near-supervised SOTA: Rapid convergence by epoch 9        |

**Key Research Finding:** An exponential performance jump occurs between **100 and 250 images** (Dice increases from 57.65% to 82.46%). At 250 images, the dataset provides sufficient pigment diversity for MobileNetV3's multi-scale skip connections to fully engage, proving that **250 samples represent the critical viability threshold** for clinical transfer learning in dermoscopy.

6. Interactive Streamlit Dashboard System

- **Executive Dashboard:** Status badge, 4-step pipeline visualizer, live validation score cards, and color-coded Dice progress bars.
- **Experiment Lab:** Hyperparameter inspection, Plotly learning curves (Data Size vs Dice/IoU), and data insights.
- **Segmentation Studio:** Sample ISIC library or upload, dynamic threshold slider ( $\theta$ : 0.10–0.90), 4-panel visualizer (Original, Heatmap, Mask, Overlay), mask export button, and side-by-side comparative panel across all 4 models.
- **Model Comparison:** Score cards, interactive **Dice Coefficient Matrix Heatmap**, side-by-side grouped bar charts, and radar fingerprints.
- **Test Evaluation:** Discovers image-mask pairs in sample\_data/ to compute live Dice, IoU, Precision, and Recall on the fly.
- **About Project:** Technical methodology, architectural diagrams, and clinical disclaimers.

7. Cloud Deployment & Academic Disclaimer

**Continuous Cloud Deployment:** The codebase is deployed on **Streamlit Community Cloud** connected to GitHub repository Gowthami1214/SkinGenAI on branch main. Dependencies in requirements.txt utilize PyTorch's CPU extra-index-url, reducing cloud image build sizes from 2.5GB to ~170MB.

**Clinical Disclaimer:** SkinGenAI is an academic research prototype designed to study few-shot transfer learning under data scarcity. Predictions are not certified by the FDA, CE, or medical regulatory bodies, and must never be used for primary diagnostic purposes without a licensed dermatologist.